

Ultimate Nutrition

CREATINE MONOHYDRATE

The major challenge for athletes and physical fitness buffs is to keep the muscles and brain energized. This is especially the case for endurance athletes who also need to recover their muscle power quickly after an intense workout and exercise.

Muscle contraction and relaxation are driven by energy in the cells provided by the compound adenosine triphosphate (ATP). Muscles have only a limited supply of ATP, however. The ATP levels are rapidly used up during exercise, particularly in short, intense bursts of anaerobic activity. Many an athletes needs immediate replenishment of energy in the muscle during demanding regimens, such as sprinting and weightlifting. In these situations, the body produces ATP: The energy currency in the cells, which powers muscle movement and control.

How much and how efficiently is ATP produced depends upon the amounts of creatine phosphate stored in the cell. Creatine is the biomolecule, which produces ATP. Creatine is indigenous to the human body, and is produced by the liver and stored in the brain, skeletal and heart muscles where it serves as a ready pool to produce ATP. Creatine is the key substance that helps speed up the ATP refueling of the muscle. More creatine in the muscle means greater energy production that allows muscle to work longer and at higher intensity.

How does creatine help the muscle perform at higher efficiency and recover faster? Creatine saturates the muscle to convert it into ATP. This conversion takes place in the absence of oxygen and is, therefore, used as an anaerobic pathway for energy.

Creatine is, however, more than just an ergogenic nutrient: It has a volumizing effect on the muscle. Cell volumizing, or cellular hydration, involves binding of water to the muscle fiber. In turn, this binding gives rise to the "pumped" look. Since water is bound inside the muscle, the athletes using creatine do not have the "puffy" appearance. In other words, the muscle looks larger but, more importantly, has improved strength and has high energy and intensity. That allows the athlete to recover faster and have more energy for each training session.

SELECTED REFERENCES

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FREQUENTLY ASKED QUESTIONS

1. What is Creatine Monohydrate Dosage?
Creatine is taken in two phases. The first phase is the so-called "loading phase" when creatine is used to load the body. Depending on the individual, it could vary between one and five days with approximately five grams of creatine is taken three to four times a day. In the "maintenance" phase following the "loading" phase, creatine levels are maintained by a dose of five to 15 grams per day in divided dosages of five gram at one sitting for up to three months.
2. Who Uses Creatine Monohydrate?
According to reports professional athletes and Olympians use creatine monhydrate to enhance their performance. Creatine is taken most commonly ingested by weightlifters and bodybuilders. Since muscle strength and endurance increases all athletic performance, athletes of all stripes benefit from creatine supplementation, including runners, swimmers, cyclists, skiers, martial artists, wrestlers, football and rugby players and tennis and basketball players.
3. Is Creatine Found in Foods?
To varying degrees creatine is found in many foods. The richest source of creatine is lean red meat and fish. A 2.2-pound steak contains roughly six grams of creatine. It should be pointed out that eating large amounts of red meat might be unhealthy.
4. Is There Creatine Deficiency?
Since creatine breaks down rapidly during the first few minutes of intense exercise, creatine deficiency may ensue. That may result in decline in muscle power and onset of fatigue during intense muscle contractions.
5. Is There Creatine Balance in the Muscle?
Yes, the muscle has creatine equilibrium. Creatine has a normal turnover rate of

2 grams per day. That is, two grams of creatine are irreversibly degraded and, in its place, equivalent amount is synthesized to replace it. There is a balance between creatine pool inside the muscle cells and creatine circulating in the blood. After the start of creatine supplementation, this balance may be slightly perturbed, but the body quickly adjusts to the new equilibrium within a few days.